Diffusion-Augmented Cross-Domain Prototypical Knowledge Distillation for Few-Shot Learning in Hyperspectral Image Classification

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***Abstract*— Cross-domain few-shot learning (CDFSL) has demonstrated remarkable new class recognition capabilities in hyperspectral image classification (HSIC) tasks. However, existing domain adaptation methods face two critical challenges in the cross-domain feature alignment process: first, the domain shift leads to misaligned feature transfer and diminished classification accuracy, and second, the intraclass feature dispersion and interclass boundary blurring in few-shot tasks result in degraded classification performance for novel classes. Moreover, the impact of redundant and noisy data on model discriminability is rarely considered in existing approaches. To solve these issues, this article proposes a cross-domain FSL HSIC method based on diffusion-augmented prototype knowledge distillation. First, we introduce a diffusion-augmented unsupervised domain adaptation pretraining framework to address the domain shift by performing a domain-adversarial (DA) denoising and reconstruction task using visible source data and masked target data. Second, our dual-branch spatial–spectral attention (DBSSA) captures global and local spectral–spatial dependencies to enhance feature representation. Then, the proposed global–local prototype knowledge distillation (GL-PKD) performs global prototype alignment while conducting local contrastive learning, addressing feature dispersion and boundary ambiguity. Finally, a dynamic learning strategy prioritizes feature alignment early, gradually strengthens classification supervision through adaptive loss weights, and incorporates a signal-to-noise ratio (SNR)-enhanced loss to effectively mitigate noise interference. The experimental results on three HSI datasets demonstrate the superiority and effectiveness of the proposed cross-domain FSL based on diffusion-augmented prototype knowledge distillation (DAPKD-CFSL).**

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***Index Terms*— Cross domain, diffusion model, few-shot learning (FSL), hyperspectral image classification (HSIC), knowledge distillation, prototypical learning.**

I. Introduction

HYPERSPECTRAL image (HSI) is a type of remote sensing images with multidimensional spatial–spectral features, which can precisely identify some targets that are difficult to distinguish in natural light images through rich spectral information [1], [2]. Therefore, HSI classification (HSIC) can be widely applied in fields such as mineral resource exploration [3], precision agricultural management [4], ecological disaster early warning [5], and military target reconnaissance [6].

Early HSIC studies mainly relied on traditional methods based on manual feature extraction, which were difficult to meet practical needs because of their dependence on domain prior knowledge and the bottleneck of high-dimensional data processing [7], [8], [9], [10], [11]. With the breakthrough of deep learning in the field of remote sensing [12], [13], especially in the direction of HSIC [14], [15]. After the successful application of stacked autoencoder (SAE) in HSIC, researchers have proposed various network architectures, such as convolutional neural network (CNN) [16], [17], [18], recurrent neural network (RNN) [19], graph convolutional network (GCN) [20], [21], and Transformer [22], [23]. These methods have significant advantages in feature representation, spectral dimensionality reduction, and generalization ability, but their performance is still limited by the amount of labeled data. Therefore, how to reduce the reliance on a large amount of labeled data without affecting classification accuracy has become a key issue in HSIC research.

To address the challenge of limited labeled data in HSIC, current research mainly focuses on data augmentation [24], [25] and pseudo-label generation [26], [27]. These methods can improve the separability of classes in high-dimensional feature spaces and increase the available training data volume. However, they carry the risk of feature space collapse due to noise propagation and may reduce model robustness and exacerbate over-fitting when overly dependent on pseudo-labels, thereby undermining the classification performance. In contrast, few-shot learning (FSL) has become the mainstream paradigm for hyperspectral few-shot classification by

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building a transferable knowledge system with limited labeled data [28], [29]. Currently, most mainstream FSL methods adopt meta-learning frameworks based on metric learning and embedding optimization. Liu et al. [30] proposed the deep FSL (DFSL) method, which constructs a metric matrix for class separability through the Euclidean distance in the metric space and combines the K-nearest neighbor (K-NN) algorithm to achieve feature matching, effectively alleviating the degradation of model performance caused by scarce labeled data. Wang et al. [31] developed the global prototype network, which uses a global representation learning strategy to construct class prototypes in the embedded feature space, significantly enhancing the model's domain adaptation ability in FSL scenarios. The method mentioned above achieves effective domain adaptation and good classification accuracy with a small number of labeled data when meta-training and meta-testing categories are identically distributed.

However, cross-domain category distribution differences lead to biases during feature alignment, significantly weakening the model's generalization ability [32]. Faced with this challenge, Li et al. [33] were the first to model the knowledge transfer between base and novel classes as a domain-shift phenomenon, achieving cross-domain feature alignment through an adversarial domain adaptation framework, thereby establishing the cross-domain FSL (CDFSL) task paradigm. Building on this foundation, researchers have proposed models such as Gia-CFSL [34], ADAFSL [35], and GLGAT-CFSL [36], which mitigate the domain-shift problem by adjusting the feature distributions between source and target domains (TDs). However, these methods exhibit certain limitations. First, during meta-training, the dominance of source-domain data leads to distribution alignment biased toward the source domain, resulting in systematic bias in target-domain predictions [37], [38], [39]. Second, these methods primarily rely on direct feature alignment techniques, neglecting the nonlinear correlations between spectra, which hinders the capture of spatial–spectral coherence in hyperspectral data and limits model performance [40], [41], [42].

To address these limitations, knowledge distillation employs a decoupling distillation strategy that splits knowledge into domain-general (spectral commonality) and domain-specific (target-domain key features). During target-domain fine-tuning, the student model learns domain-general knowledge from the teacher while acquiring target-specific features, reducing over-reliance on source-domain features [43], [44]. Feng et al. [45] proposed a CDFSL classification framework based on decoupled knowledge distillation. It applies knowledge distillation strategies to prioritize key target-domain features and reduce the impact of biased meta-knowledge. It also minimizes distances between similar spectral bands via feature maps, enhancing the model's ability to capture spectral correlations.

However, the existing cross-domain knowledge distillation methods have critical limitations. The teacher network is only pretrained on the source domain, neglecting the adaptation process to the target domain, which makes it difficult for it to distinguish domain-related noise from domain-invariant semantic features in cross-domain tasks [46], [47]. This source-domain feature-dominated pattern intensifies the model's excessive reliance on source-domain supervision signals, leading to significant shifts between domains. Such domain shifts not only result in class dispersion and boundary ambiguity in the feature space of the target domain, but also give rise to cross-domain noise interference issues. Although data augmentation can partially alleviate domain shift, the spectral shift between the source and target domains causes the class-relevant knowledge extracted by the model to fail on the target domain. It is worth noting that the methods of unsupervised domain adaptation aim to extract domain-invariant features, alleviate spectral shift, and classify target samples using labeled source data and unlabeled target data [48], [49].

To address the aforementioned challenges, this study proposes a cross-domain FSL based on diffusion-augmented prototype knowledge distillation (DAPKD-CFSL) for HSIC method. First, inspired by diffusion model, we design a diffusion-augmented pretraining (DA-PT) module grounded in unsupervised domain adaptation to alleviate the domain shift caused by source-domain features. By jointly training visible source data and masked target data via denoising and reconstruction tasks along with domain adversarial (DA) constraints, the teacher network extracts highly robust cross-domain features, focuses on mining domain-invariant features, and suppresses the interference of interdomain noise during key feature learning. Second, the dual-branch spatial–spectral attention (DB-SSA) module captures both spectral global dependencies and spatial local details, providing enhanced discriminative representations of spectral bands and key spatial regions for subsequent feature alignment. Subsequently, to tackle cross-domain feature dispersion and boundary ambiguity, we propose a global–local prototype knowledge distillation (GL-PKD) strategy. This strategy leverages the prior knowledge of the teacher model to guide the optimization of the target domain while reducing reliance on source-domain supervision signals. Specifically, it extracts global category prototypes from the target domain and achieves cross-domain category center alignment by minimizing the distance between the student and teacher model prototypes. Simultaneously, it performs local feature contrastive learning through historical student models, constructs positive and negative sample pairs, and maximizes the similarity of positive samples to enhance intraclass consistency and interclass separability while completing global feature alignment. Finally, a dynamic learning is introduced to balance knowledge transfer and classification accuracy by dynamically adjusting loss weights, emphasizing feature alignment in the early stages and progressively strengthening classification supervision later. Additionally, a signal-to-noise ratio (SNR)-enhanced loss is incorporated to further alleviate residual noise interference effectively.

The primary contributions of this article are summarized as follows.

1) We proposed a novel CDFSL framework DAPKD-CFSL, which combines an unsupervised domain adaptation pretraining module with a prototype knowledge distillation strategy to suppress the knowledge biased toward the source domain and reduce the interclass

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confusion and noise interference in the target domain, ultimately enhancing the accuracy of CDFSL HSIC.

2) The DA-PT module for the first time combines diffusion models with DA learning to address the challenges in domain-invariant feature learning and suppress domain shift. By jointly optimizing the diffusion denoising loss and the DA loss, it enables the encoder to extract highly robust cross-domain features from noisy data, while effectively alleviating the feature distribution differences between the source and target domains.
3) To address the limitations of traditional knowledge distillation methods in aligning class features, the GL-PKD module first aligns the class centers of the teacher and student models through minimizing global prototype distances. Meanwhile, a local contrastive learning is introduced to enhance class boundary sensitivity and promote the aggregation of samples within the same class, effectively overcoming the problems of class dispersion and boundary ambiguity that traditional methods encounter in the domain feature space.
4) To suppress the interference caused by cross-domain noise and redundant bands during the extraction of discriminative meta-knowledge, we propose a DB-SSA module. This module performs dynamic feature selection from both spatial and spectral dimensions, focusing on cross-domain discriminative bands and spatial regions to highlight key features, and reduces the difficulty of extracting discriminative features.

The subsequent structure of this article is as follows. Section II provides an overview of the concepts of CDFSL, knowledge distillation, and diffusion model augmentation. Section III offers a comprehensive introduction to the DAPKD-CFSL method. Section IV analyzes the ablation and comparison experiments of the DAPKD-CFSL results. Discussions on the number of training samples, feature extraction capabilities, hyperparameters analysis, and computational complexity are presented in Section V. Conclusions are drawn in Section VI.

II. Related Work

This section introduces some works related to CDFSL, knowledge distillation, and diffusion augmentation.

A. Cross-Domain FSL

CDFSL aims to perform HSIC by integrating a few labeled samples from the target domain with abundant labeled data from the source domain. This approach improves target-domain accuracy with limited samples by jointly training on both domains, leveraging source-domain data to construct transferable meta-knowledge. Notably, source and target domains may have different classes, yet the general knowledge learned from the source domain aids feature extraction and classification in the target domain [50], [51]. This enables FSL methods to achieve promising results in scenarios with scarce labeled samples and domain differences [28], [52], [53].

In FSL tasks for HSIC, the source dataset D_s contains a large number of labeled samples across C_s classes, while the target dataset D_t contains only a small number of labeled samples across C_t classes. To ensure diversity in the training samples, the number of classes in the source domain C_s should be equal to or more than the number of classes in the target domain C_t. This facilitates the transfer of meta-knowledge from the source domain to the target domain, thereby enabling effective classification in the target domain [54].

Overall, cross-domain FSL aims at solving the problem of domain transformation between the source and target domains. However, due to the dominant position of source-domain data in the meta-training stage, distribution alignment is biased toward the source domain, thereby restricting the model's performance. To address this issue, this article proposes a method centered on knowledge distillation, which guides the student model to focus on the key features of the target-domain data and gradually reduces the influence of biased meta-knowledge, thereby enhancing the model's cross-domain adaptability.

B. Knowledge Distillation

Knowledge distillation transfers the knowledge of large models to efficient small models through a teacher–student architecture [55], [56]. By training the teacher model on the labeled data, soft labels encoding the similarity of categories are generated. Subsequently, the student model improves its generalization ability through dual supervision learning, which involves imitating the soft label distribution produced by the teacher model and matching the labels of the original dataset by minimizing the Kullback–Leibler (KL) divergence between their outputs

$$\mathcal{L}_{\mathrm{KL}}\left(\mathcal{N}_S, x, \hat{y}\right) = \sum_{x} \hat{y} \log \frac{\hat{y}}{\mathcal{N}_S(x)} \tag{1}$$

where $\mathcal{N}_S$ represents the student model, x is the input, and $\hat{y}$ is the predicted output.

In the field of HSIC, due to limited labeled data, knowledge distillation has shown unique advantages. Yue et al. [57] proposed an adaptive framework leveraging self-supervised learning to leverage unlabeled data. Zhao et al. [58] developed a lifelong learning strategy that continuously updates spectral–spatial feature distillation weights for a universal classification model. Chi et al. [59] integrated self-supervision into knowledge distillation to enhance feature discriminability. Notably, Feng et al. [45] introduced decoupled meta-knowledge distillation, jointly optimizing knowledge transfer and domain-shift correction to improve cross-domain performance. Additionally, Zhang et al. [60] proposed an ensemble network-based distillation method to address label noise in HSIC.

Although knowledge distillation works well in cross-domain HSIC, a teacher network trained only on the source domain fails to adapt to the target domain and struggles to distinguish domain noise from invariant semantic features in hyperspectral data. Additionally, traditional knowledge distillation overlooks intraclass structures, causing interclass overlap and fine-grained feature loss. In this article, we use

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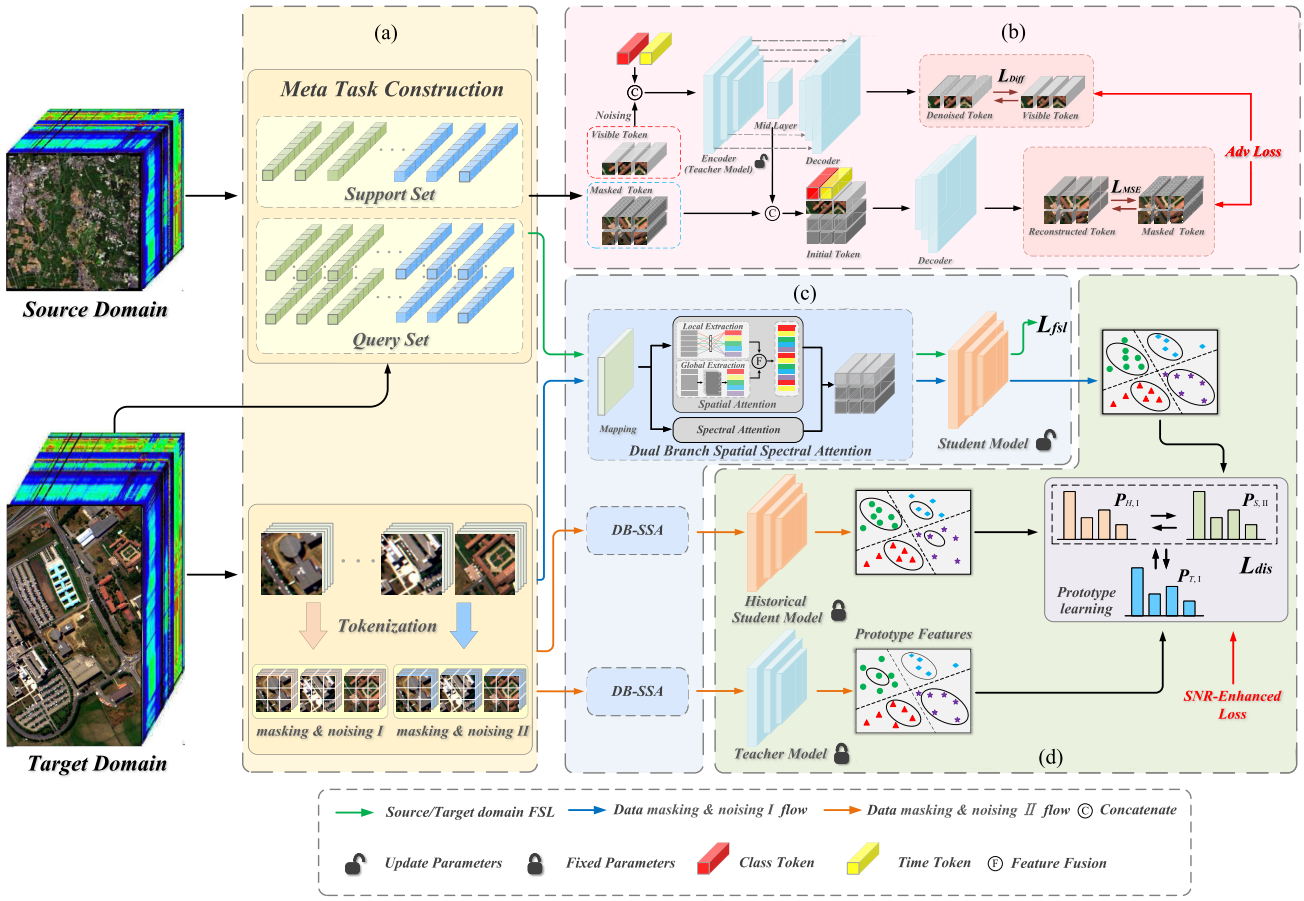


Fig. 1. Flowchart of the DAPKD-CFSL method. In the data preparation stage, both the source data and target data will be structured into meta tasks, and a part of the target data will undergo masking and noising. During the DA-PT stage, DA training is conducted concurrently with denoise and reconstruction tasks. During the meta-knowledge extraction stage, the DB-SSA module enhances the representation of discriminative bands and key regions, while the student model acquires meta-knowledge from the data using the FSL strategy. In the prototype knowledge distillation stage, the teacher network extracts class prototypes from target-domain data to achieve global prototype alignment, while the historical student network enforces local feature consistency through contrastive learning. Meanwhile, the global prototype distillation loss and local contrastive loss are adaptively weighted and fused. (a) Data preparation. (b) DA-PT. (c) Meta-knowledge extraction. (d) Prototype knowledge distillation.

a diffusion-enhanced pretrained model as the teacher network and address these issues by integrating prototype learning and contrastive learning into knowledge distillation.

C. Diffusion Augmentation

Diffusion models, originally designed for image generation, are increasingly applied to hyperspectral representation learning due to their denoising capabilities. However, the current single pretraining task suffers from limited feature extraction, which can be improved by adopting the multistep prediction strategy of diffusion models based on a parametric Markov chain framework. To address this, we propose a new pretraining paradigm from the perspective of representation learning to enhance the semantic understanding of HSIs. Preechakul et al. [61] introduced the diffusion autoencoder (DAE), which separates deep semantic features for classification and decodable features for high-precision reconstruction. Building on this, Li et al. [62] proposed the diffusion enhanced masked autoencoder (DEMAE), which integrates the strengths of diffusion models and masked autoencoders to significantly improve feature representation under self-supervised learning.

Although existing diffusion-augmented methods have integrated diffusion models with masked autoencoder, they remain constrained by single-domain small-sample scenarios in cross-domain transfer tasks. This article proposes leveraging the robust prior features generated through the diffusion process to facilitate domain-invariant feature extraction. Additionally, DA training is utilized to refine feature boundaries, mitigate domain distribution discrepancies, and thus significantly improve feature discriminability in cross-domain tasks.

III. Methods

The overall flowchart of the proposed DAPKD-CFSL is illustrated in Fig. 1. First, meta-tasks are constructed for both the source- and target-domain data, and masks and noise are introduced to data augment in the data preparation phase. Subsequently, in the DA-PT stage, cross-domain robust features are learned via denoise and reconstruction tasks. A domain discriminator is embedded to perform DA training in the middle layer, aligning the feature distributions between the source and target domains. In the meta-knowledge extraction stage, the DB-SSA module processes all data to achieve

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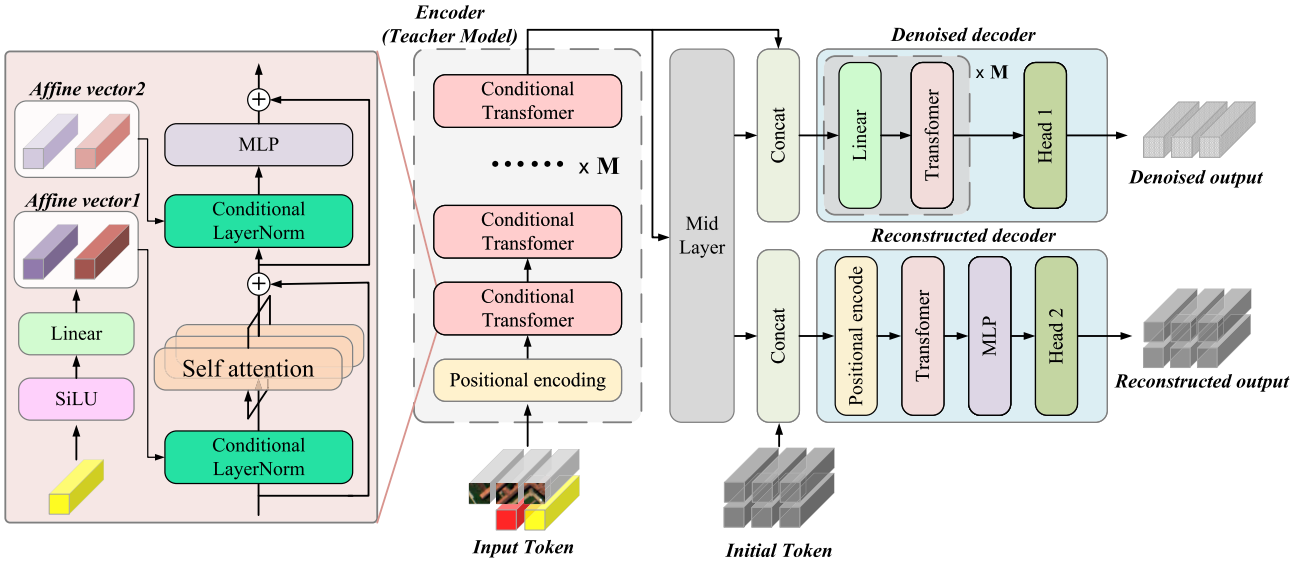


Fig. 2. Structure diagram of the DA-PT module. The visible samples carry time tokens and class tokens, which are first processed by the conditional transformer as an encoder to extract features. Subsequently, the features pass through the domain discriminator in the intermediate layer to achieve DA training. Finally, the denoising and reconstruction tasks are accomplished by the mask decoder and the noise decoder. It is worth noting that in subsequent tasks, the conditional transformer part will be used, respectively, as the teacher model and the student model.

feature decoupling and capture global and local dependencies in spectral and spatial dimensions, enhancing the representation of discriminative bands and key regions. Next, the FSL strategy is applied to the student model to extract shared meta-knowledge. During this phase, predictions on the same target-domain data are made by the student model, teacher model, and historical student model. In the GL-PKD stage, the teacher model extracts class prototypes from the target domain to achieve global prototype alignment, while the historical student network enforces local feature consistency through contrastive learning. Finally, the global prototype distillation loss and local contrastive loss are adaptively weighted and fused, improving classification accuracy in the target domain and enhancing robustness to domain shift and noise interference.

A. Data Preparation

In data preparation stage, we formally denote the input HSIs from the source domain and the target domain as $\mathbb{R}^{h\times w\times \mathrm{ch}_s}$ and $\mathbb{R}^{h\times w\times \mathrm{ch}_t}$, respectively, where h and w represent the HSI's spatial dimensions of height and width, and ch_s and ch_t correspond to the HSI's spectral dimensions in the source and target domains, respectively. First, we extract $x_{s/t} \in p \times p \times \mathrm{ch}_{s/t}$ neighborhood data cubes centered at each pixel from the HSIs represented by $\mathbb{R}^{h\times w\times ch_{s/t}}$, where p is the size of patches. Then, to address the challenge of varying spectral resolutions due to different sensors in the source and target domains, a mapping layer converts $p \times p \times \mathrm{ch}_s$ from the source domain and $p \times p \times \mathrm{ch}_t$ from the target-domain cubes into unified cubes $p \times p \times B$ through spectral alignment, where B denotes the spectral dimension. Finally, N samples will be selected from C_s classes that are divided into support sets $S_s = \{(x_i, y_i)\}_{C_s\times N_s}$ and query sets $Q_s = \{(x_j, y_j)\}_{C_s\times N_q}$, where N_s and N_q represent the number of support sets and query sets, respectively. The support set serves as a dataset for training, while the query set is utilized to evaluate the model's performance.

In classical knowledge distillation, temperature adjustment is used to modulate the distribution of soft labels. However, the selection of the optimal temperature often relies heavily on human experience. To address this issue, the proposed method employs strong masking and noising to enable the student model to learn robust features, while the teacher model generates supervisory signals from weakly masked and noised images. This approach not only enhances the knowledge diversity for the student model, but also focuses on image feature learning, thereby reducing its reliance on temperature tuning. Given labeled sample data $x_i \in \mathbb{R}^{p\times p\times B}$, applying masks and noise of varying degrees produces two augmented versions x_{I} and x_{II} with the same label y_i.

B. Diffusion-Augmented Pretraining Stage

In the DA-PT pretraining stage, the DA-PT module enables the model to learn robust cross-domain features from both visible source data and masked target data, thereby reducing the domain shift between the source and target domains. This is achieved through joint training on DA tasks, denoising tasks, and reconstruction tasks.

First, let HSI patch $x_i \in \mathbb{R}^{p\times p\times B}$ be sampled from visible source data and masked target data. Then, x_i is divided into a visible subset x_{vis} and a masked subset x_{mask} through randomized masking operations at a predetermined ratio. During this process, the central pixels of x_i are specifically retained in x_{vis} to preserve critical features. Denoising tasks are performed on x_{vis}, while feature reconstruction tasks are executed on x_{mask}, thereby enhancing the model's representational learning capabilities. Noise addition operation is performed on x_{vis}. The forward process of the diffusion model employs a Markov chain for modeling. Note that the initial state x_0^{vis} is no noise,

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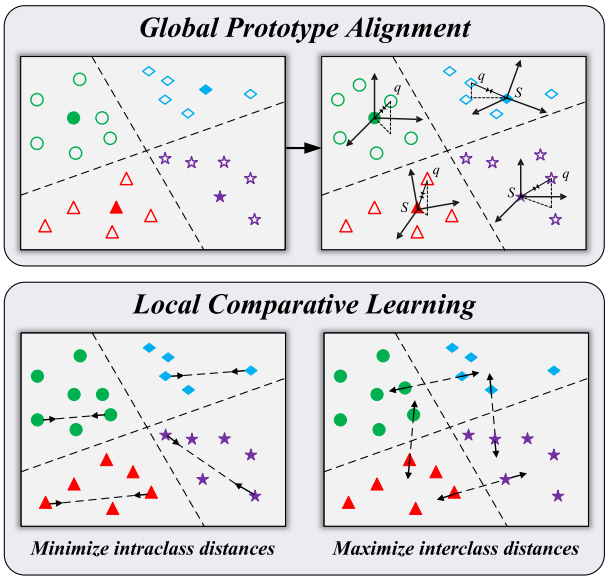


Fig. 3. Schematic of prototype alignment and comparative learning. The student model gradually aligns the prototype distribution by minimizing the distance from the teacher prototypes, thereby constructing a class-discriminative structure consistent with the teacher model. Meanwhile, local contrastive learning achieves compact intraclass distribution and significant interclass separation in the feature space by maximizing the similarity between positive sample pairs and minimizing the similarity between negative sample pairs.

TABLE I
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN THE CHIKUSEI DATASET

No.	Land cover class	Pixels
1	Water	2845
2	Bare soil(school)	2859
3	Bare soil(park)	286
4	Bare soil(farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice field(grown)	13369
10	Rice field(first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade(non-dark)	1220
14	Manmade(dark)	7664
15	Manmade(blue)	431
16	Manmade(red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
Total		77592

and x_t^{vis} denotes the data obtained by weighting the signal and noise. Phased noise addition operation is implemented on x_{vis}, where the forward diffusion process is modeled via a Markov chain. Specifically, x_0^{vis} denotes the original noise-free state, while x_t^{vis} represents the data state after t-step noise injection, where $t \in [1, T]$, with T is a hyperparameter representing the total number of noise levels. Therefore, the forward noise addition is given as follows:

$$x_t^{\text{vis}} = \sqrt{\alpha_t} x_{t-1}^{\text{vis}} + \sqrt{1-\alpha_t}\epsilon \tag{2}$$

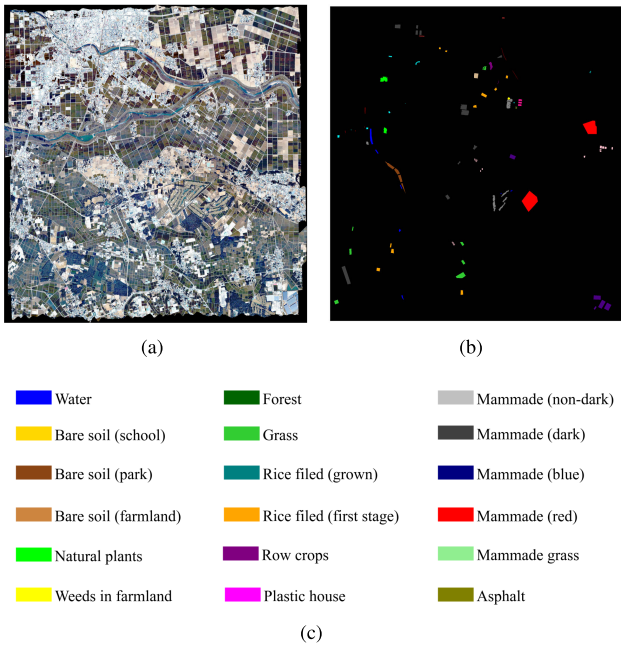


Fig. 4. Chikusei dataset. (a) Pseudo-color composite image. (b) Ground-truth map. (c) Corresponding color labels.

where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, $\alpha_t = \prod_{i=1}^{t}(1-\beta_i)$, with $\beta_t \in (0, 1)$ being predefined noise schedule parameters, and $\mathbf{I}$ is a unit matrix.

Second, the two specialized tokens are concatenated to the input sequence: a class token x_{class} for classification tasks and a time embedding token x_{time} encoding diffusion process. The class token x_{class} is randomly initialized, whereas the time token x_{time} is parameterized by the noise level t using the sinusoidal positional encoding formula

$$x_{\text{time}} = \left[\sin(t \times \exp(i \times k)),\ \cos(t \times \exp(j \times k))\right] \tag{3}$$

$$k = \frac{-\log(10000)}{d_{\text{time}}/2 - 1} \tag{4}$$

where d_{time} denotes the embedding feature dimension of the time token x_{time}, and $i \in [0, d_{\text{time}}/2 - 1]$ and $j \in [d_{\text{time}}/2, d_{\text{time}} - 1]$. Subsequently, x_{time} undergoes a two-layer multi-layer perceptron (MLP) embedding and is then incorporated into the sequence.

Third, since the conditional transformer architecture exhibits significant advantages in image generation, this architecture encodes the category labels as conditional vectors and integrates them into the layer normalization module through affine transformation. Fig. 2 achieves adaptive feature calibration for the input patches at varying noise levels by establishing a relationship between the time token x_{time} and the affine parameters of layer normalization. The conditional encoding process can be formally expressed as

$$z_{(0)}^{\text{en}} = \left[x_{\text{class}}; x_{\text{time}}; x_{1-t}^{\text{vis}}E; \cdots; x_{N-t}^{\text{vis}}E\right] + E_{\text{pos}}^{\text{vis}} \tag{5}$$

$$z_{(m)}^{\text{en}\prime} = \text{Att}\left(\text{cNorm}\left(z_{(m-1)}^{\text{en}}, x_{\text{time}}\right)\right) + z_{(m-1)}^{\text{en}} \tag{6}$$

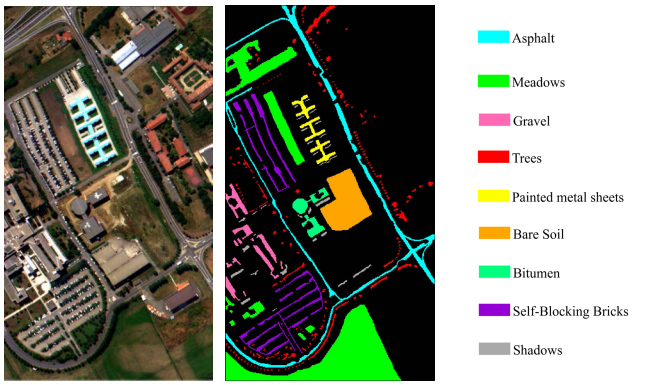
$$z_{(m)}^{\text{en}} = \text{FFN}\left(\text{cNorm}\left(z_{(m)}^{\text{en}\prime}, x_{\text{time}}\right)\right) + z_{(m)}^{\text{en}\prime} \tag{7}$$

where $x_{(p-t)}^{\text{vis}}$ denotes the pth input token in the visible input x_{vis}, E represents the linear embedding layer, and $E_{\text{pos}}^{\text{vis}}$ indicates the positional embeddings for visible tokens. The index $m \in [1, M]$, where M is the number of layers in

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TABLE II
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN PAVIAU DATASET

No.	Land cover class	Pixels
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Sheets	1345
6	Bare soil	5029
7	Bitumen	1330
8	Bricks	3682
9	Shadow	947
Total		42776



(a) (b) (c)

Fig. 5. PaviaU dataset. (a) Pseudo-color composite image. (b) Ground-truth map. (c) Corresponding color labels.

TABLE III
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN SALINAS DATASET

No.	Land cover class	Pixels
1	Brocoli_green_weeds_1	2009
2	Brocoli_green_weeds_2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807
Total		54129

the encoder, while cNorm$(\cdot)$ implements the conditional layer normalization operator.

The middle layer connecting the encoder and decoder can be represented by the following formula:

$$z_{\text{mid}} = \text{LayerNorm}\left(z^{\text{en}}_{(m)}\right) \tag{8}$$

where LayerNorm$(\cdot)$ represents the layer normalization operation, and $z^{\text{en}}_{(m)}$ indicates the intermediate feature representation produced by the encoder module.

Then, the domain discriminator is introduced in the middle layer connecting the encoder and decoder that can be represented by the following formula, which is constructed in combination with the gradient reversal layer (GRL). During the loss backpropagation stage, the gradient of the domain discriminator is reversed, thereby driving the encoder network to learn domain-invariant features generated by the encoder

$$z_f = \text{Discriminator}(z_{\text{mid}}) \tag{9}$$

where Discriminator$(\cdot)$ denotes the domain discriminator, and the adversarial loss function is given as follows:

$$\mathcal{L}_{\text{adv}} = -\mathbb{E}x_s\left[\log D(f_s)\right] - \mathbb{E}x_t\left[\log(1 - D(f_t))\right] \tag{10}$$

where $f_{s/t} = \mathcal{E}(x^{\text{vis}}_{s/t})$ denotes the domain-invariant features extracted by the encoder. $D(\cdot)$ is the domain discriminator, and $\mathbb{E}$ represents the expectation.

Furthermore, the conditional transformer-based encoder and diffusion decoder are utilized to conduct the reverse denoising process. The diffusion decoder can be represented by the following formula:

$$z^{\text{de}}_{(0)} = z_{\text{mid}} \tag{11}$$

$$z^{\text{de}\prime}_{(m)} = \text{Att}\left(\text{Norm}\left(z^{\text{de}}_{(m-1)}\right)\right) + z^{\text{de}}_{(m-1)} \tag{12}$$

$$z^{\text{de}}_{(m)} = \text{FFN}\left(\text{Norm}\left(z^{\text{de}\prime}_{(m)}\right)\right) + z^{\text{de}\prime}_{(m)}. \tag{13}$$

The reverse process, modeled as a Markov chain and denoted by $p_\theta(x^{\text{vis}}_{t-1} \mid x^{\text{vis}}_t)$, aims to gradually eliminate the Gaussian random noise introduced during the forward diffusion process. Here, θ represents the learnable parameters of the denoising model. For the denoised output x^{vis}_{t-1} generated from x^{vis}_t, we require the reverse process to effectively remove the additive noise while preserving data consistency, which minimizing the distributional divergence between the learned reverse process $p_\theta(x^{\text{vis}}_{t-1} \mid x^{\text{vis}}_t)$ and the target distribution $q(x^{\text{vis}}_{t-1} \mid x^{\text{vis}}_t)$. Hence, we enforce variance equivalence between p_θ and q at each timestep t, and derive $q(x^{\text{vis}}_{t-1} \mid x^{\text{vis}}_t)$ through Bayesian inference

$$\begin{aligned}
&q\left(x^{\text{vis}}_{t-1} \mid x^{\text{vis}}_t, x^{\text{vis}}_0\right) \\
&= \frac{q\left(x^{\text{vis}}_t \mid x^{\text{vis}}_{t-1}, x^{\text{vis}}_0\right) q\left(x^{\text{vis}}_{t-1} \mid x^{\text{vis}}_0\right)}{q\left(x^{\text{vis}}_t \mid x^{\text{vis}}_0\right)} \\
&\propto \exp\left\{-\frac{1}{2}\left[\frac{-2\sqrt{\alpha_t}x^{\text{vis}}_t x^{\text{vis}}_{t-1} + \alpha_t\left(x^{\text{vis}}_{t-1}\right)^2}{1-\alpha_t}\right.\right. \\
&\qquad \left.\left. + \frac{\left(x^{\text{vis}}_{t-1}\right)^2 - 2\sqrt{\bar{\alpha}_{t-1}}x^{\text{vis}}_0 x^{\text{vis}}_{t-1}}{1-\bar{\alpha}_{t-1}}\right]\right\} \\
&\propto \mathcal{N}\left(x^{\text{vis}}_{t-1}; \frac{\sqrt{\alpha_t}\left(1-\bar{\alpha}_{t-1}\right)x^{\text{vis}}_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)x^{\text{vis}}_0}{1-\bar{\alpha}_t}\right. \\
&\qquad \left. \frac{(1-\alpha_t)\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t}\mathbf{I}\right)
\end{aligned} \tag{14}$$

where the noise term $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ is sampled from a standard normal distribution, with $\mathbf{I}$ is a unit matrix.

Finally, the canonical diffusion framework employs the loss function $L_{\text{simple}}(\theta) = \|\epsilon_\theta(\mathbf{x}_t, t) - \epsilon\|_2^2$ to optimize model parameters θ for image generation, where $\epsilon_\theta : \mathbb{R}^d \times \mathbb{N} \rightarrow$

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$\mathbb{R}^d$ denotes the model's parameterized estimate of the noise component, and $\mathbf{x}_t$ represents the diffused sample at timestep t. Therefore, two decoders are constructed using the standard Transformer, respectively, for the HSI denoising and reconstruction tasks

$$\mathcal{L}_{\text{diff}} = \left\|\hat{x}_\theta^{\text{vis}}\left(x_t^{\text{vis}}, t\right) - x_0^{\text{vis}}\right\|_2^2 \tag{15}$$

$$\mathcal{L}_{\text{mse}} = \left\|\hat{x}_\theta^{\text{mask}}\left(x_t^{\text{mask}}, t\right) - x_0^{\text{mask}}\right\|_2^2 \tag{16}$$

where $\hat{x}_\theta^{\text{vis}}(x_t^{\text{vis}}, t)$ represents the output of the diffusion branch, and $\hat{x}_\theta^{\text{mask}}(x_t^{\text{mask}}, t)$ is the output of the MAE branch.

The pretraining loss function $\mathcal{L}_{\text{pretrain}}$ is calculated using the following loss function:

$$\mathcal{L}_{\text{pretrain}} = \mathcal{L}_{\text{diff}} + \mathcal{L}_{\text{mse}} + \mathcal{L}_{\text{adv}}. \tag{17}$$

C. Meta-Knowledge Extraction

To enhance feature representation in CDFSL for HSIC, we designed a DB-SSA module. In this module, the input HSI patches $x_i \in \mathbb{R}^{p\times p\times B}$ were initially embedded into a spatial–spectral embedding metric space, forming embedded features.

In the DB-SSA module, the spatial branch uses 2-D convolution kernels to extract spatial context features $f_s \in \mathbb{R}^{p\times p}$, while the spectral branch captures spectral sequence features $f_c \in \mathbb{R}^B$ through 1-D convolution layers. Subsequently, a 3-D average pooling operation is introduced to compress spatial dimensions and generate local channel weights $w_{\text{local}} \in \mathbb{R}^B$. The learned weights are elementwise multiplied with the input features F_c, generating the intermediate representation $f_{\text{local}} = f_c \odot w_{\text{local}}$. Then, the $f_c \in \mathbb{R}^B$ undergoes secondary 3-D average pooling, followed by two sequential $1 \times 1 \times 1$ convolutional operations: the first convolution compresses channel dimensionality to C/r (where r denotes the reduction ratio), while the second convolution reconstructs the original channel dimensions. This processing flow ultimately generates global channelwise attention weights w_{global} through sigmoid activation. These weights are applied to f_c results in f_{global}. Finally, a sigmoid activation is applied to the spatial branch F_s and combined with $f_c = f_{\text{global}} \odot f_{\text{local}}$, resulting in the output feature $f' = f_s \odot f_c$.

Then, we adopt the pretrained encoder as the feature extractor. By implementing FSL in both the source and target domains, the model can effectively achieve cross-domain knowledge transfer, thereby enhancing the recognition ability of new categories in the target domain. Taking the FSL training in the source domain as an example, during the training process, N classes are randomly selected from N_s source classes to construct a training episode. Each training episode completes few-shot classification by calculating the Euclidean distance between the feature embeddings of the query set and the category prototypes. Finally, the distance is mapped to the prediction probability of the query sample x_i based on the softmax function, and the formula is given as follows:

$$P(y_i = k \mid x_i \in Q_s) = \text{Softmax}\left(-G_E\left(f_i^s, f_k\right)\right) \tag{18}$$

where f_i^s corresponds to the embedded representation of query set samples, characterizing class-discriminative features within the latent space. The prototype vector f_k is the centroid of support set samples from the kth classes, geometrically positioning each class. The metric function G_E computes Euclidean distances between feature vectors. $x_i \in Q_s$ denotes an unlabeled query instance requiring classification, while y_i represents its ground truth.

The FSL loss function in the source domain computed from the predicted probability of query set instances is formally defined as

$$\mathcal{L}_{\text{fsl}}^s = \frac{\sum_{(x,y\in Q_s)}(-\log P(y_i = k \mid x_i \in Q_s))}{|S_s| \times |Q_s|} \tag{19}$$

where S_s represents the size of the support set, and Q_s represents the size of the query set.

Similarly, the FSL loss for the query samples in the target domain can be formulated as

$$\mathcal{L}_{\text{fsl}}^t = \frac{\sum_{(x,y\in Q_t)}(-\log P(y_i = k \mid x_i \in Q_t))}{|S_t| \times |Q_t|}. \tag{20}$$

The total FSL loss can be represented as

$$\mathcal{L}_{\text{fsl}} = \mathcal{L}_{\text{fsl}}^s + \mathcal{L}_{\text{fsl}}^t. \tag{21}$$

D. Prototype Knowledge Distillation

The core objective of CDFSL is to extract and debias meta-knowledge. To address the distribution alignment bias toward the source domain caused by its data dominance, we aim to reduce systematic bias in target-domain predictions and promote sustainable knowledge interaction between the source and target domains. The proposed DAPKD-CFSL method introduces a prototype knowledge distillation strategy to facilitate autonomous learning in the student model through teacher model guidance for the downstream task. Based on this strategy, the teacher network extracts target-domain category prototypes to achieve global feature alignment, guiding the student model to learn core domain knowledge while expanding interclass feature distances. The student model further reduces intraclass variations through historical model contrastive learning, enabling efficient knowledge extraction and precise information filtering under teacher model guidance. The schematic of prototype alignment and comparative learning is shown in Fig. 3.

The proposed GL-PKD framework comprises three core components: the student model $\mathcal{N}_S$, the teacher model $\mathcal{N}_T$, and the historical student model $\mathcal{N}_H$. Within this framework, the student model $\mathcal{N}_S$'s core function involves applying the FSL strategy to distill discriminative meta-knowledge from cross-domain data. Simultaneously, considering that the pretrained model can extract features from HSI samples with masks and noise, strong masking and noise augmentation are applied to the target-domain samples during the training of the student model $\mathcal{N}_S$ to leverage the FSL strategy to extract discriminative meta-knowledge, while weak masking and noise augmentation are implemented for the target-domain samples in the training of the teacher model $\mathcal{N}_T$ and historical student models $\mathcal{N}_H$, thereby enhancing the likelihood of achieving superior classification accuracy during the knowledge distillation stage.

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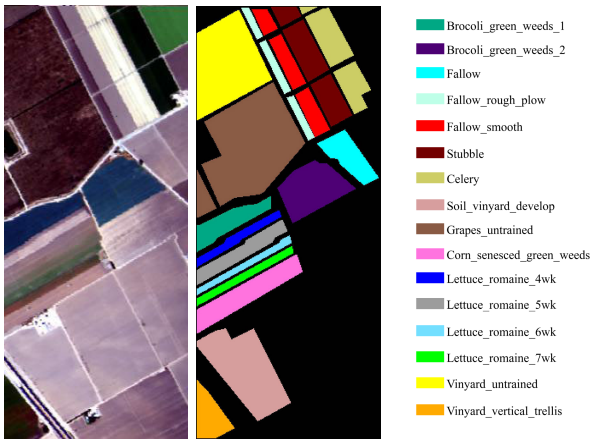


Fig. 6. Salinas dataset. (a) Pseudo-color composite image. (b) Ground-truth map. (c) Corresponding color labels.

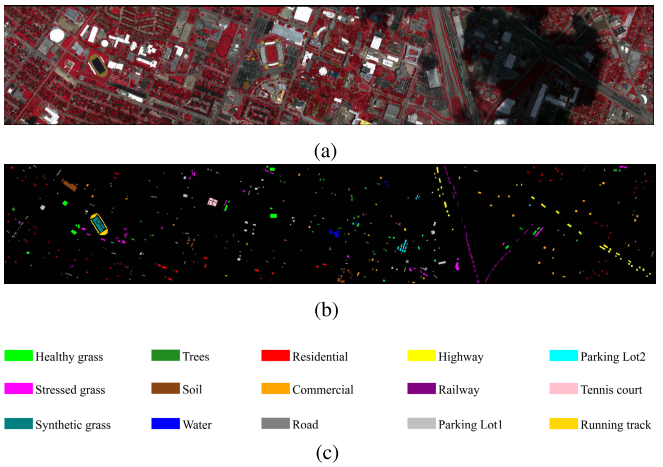


Fig. 7. Houston dataset. (a) Pseudo-color composite image. (b) Ground-truth map. (c) Corresponding color labels.

TABLE IV
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN HOUSTON DATASET

No.	Land cover class	Pixels
1	Healthy grass	1251
2	Stressed grass	1254
3	Synthetic grass	697
4	Trees	1244
5	Soil	1242
6	Water	325
7	Residential	1268
8	Commercial	1244
9	Road	1252
10	Highway	1227
11	Railway	1235
12	Parking Lot1	1233
13	Parking Lot2	469
14	Tennis court	428
15	Running track	660
Total		15029

Specifically, the embedding features f_{I}' and f_{II}', respectively, represent the weakly enhanced and strongly enhanced versions of the same image patch. After the feature representations are enhanced by the DB-SSA module, the feature f_{II}' is used in the $\mathcal{N}_S$ to obtain feature $z_{s,\mathrm{II}}$, and the feature f_{I}' is, respectively, used in the $\mathcal{N}_H$ and $\mathcal{N}_T$ to obtain features $z_{h,\mathrm{I}}$ and $z_{t,\mathrm{I}}$. $\mathcal{N}_T$ and $\mathcal{N}_S$ maintain the same initial parameter configuration during training and adopt a consistent training strategy. After each iteration, the previous generation of $\mathcal{N}_S$ is archived as $\mathcal{N}_H$ and participates in feature extraction for the subsequent training stage alongside the current model.

1) Global Prototype Alignment: The teacher model extracts class prototypes from the target-domain data, serving as discriminative reference points for that class in the feature space. The student model gradually aligns its prototype distribution by minimizing the distance to the teacher prototypes, constructing a class discrimination structure consistent with that of the teacher model. This prototype-driven transfer strategy achieves cross-domain feature alignment and conveys discriminative meta-knowledge.

Specifically, the teacher model $\mathcal{N}_T$ encodes the target-domain embedding feature f_{I}' and groups the samples by class based on the pseudo-labels it predicts. Then, the class prototype vectors for each class are calculated through the grouping

$$p_T^k = \frac{1}{N_k} \sum_{i=1}^{N_k} \mathcal{N}_T\left(f_{\mathrm{I}i}'^k\right) \tag{22}$$

where N_k represents the number of samples of the pseudo-labels belonging to class k. Similarly, the student model $\mathcal{N}_S$ extracts features from the same target-domain samples and calculates the student prototype p_S^k.

Finally, minimize the cosine similarity loss between the teacher and student prototypes

$$\mathcal{L}_{\text{global}} = \sum_{k=1}^{K} \left(1 - \frac{p_T^k \cdot p_S^k}{\left\|p_T^k\right\| \left\|p_S^k\right\|}\right). \tag{23}$$

2) Local Comparative Learning: Local contrastive learning drives the student model to learn local feature representations with class discrimination by constructing the contrast of feature similarity between positive and negative sample pairs. This method achieves compact intraclass distribution and interclass separation in the feature space by maximizing the similarity of positive sample pairs and minimizing that of negative sample pairs, thereby enhancing the model's ability to capture class boundaries.

Specifically, for the embedding features f_{I}' and f_{II}', the student model $\mathcal{N}_S$ generates a output feature $f_S(x_i)$, and the output feature of the historical student model $\mathcal{N}_H$ is $f_H(x_i)$. These two form a positive sample pair (f_S, f_H). The feature set $f_H(x_j)_{j\neq i}$ generated by the remaining samples in the same training batch is used as negative samples to participate in contrastive learning. Then, maximize the similarity of positive sample pairs while suppressing that of negative sample pairs

$$\mathcal{L}_{\text{local}} = \left(-\log \frac{\exp(f_S \cdot f_H/\tau)}{\sum_{j=1}^{N} \exp\left(f_S \cdot f_j^H/\tau\right)}\right) \tag{24}$$

where τ is the temperature coefficient.

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The loss function of GL-PKD can be expressed as

$$\mathcal{L}_{\mathrm{PKD}} = \mathcal{L}_{\mathrm{CE}} + \lambda_1 \mathcal{L}_{\mathrm{global}} + \lambda_2 \mathcal{L}_{\mathrm{local}} \tag{25}$$

where $\mathcal{L}_{\mathrm{CE}}$ is the classification loss function, and λ_1 and λ_2 are the weight parameters that are dynamically adjusted during training. In the initial stage, the focus is on global distribution alignment, while in the later stage, the emphasis shifts to enhancing local feature refinement.

Then, in order to effectively reduce the interference of redundant and noisy data on model performance by assigning greater weights to high SNR samples and imposing smaller penalties on low SNR samples, the SNR-Enhanced loss function is formulated as follows:

$$\mathcal{L}_{\mathrm{SNR}}\left(y_i, \hat{y}_i\right) = -\log\left(\frac{P(y_i)}{P\left(y_i - \hat{y}_i\right)}\right) \tag{26}$$

where y_i denotes the ground truth, $\hat{y}_i$ represents the predicted result, $P(y_i) = \mathbb{E}[|y_i|^2]$ quantifies the power of the clean signal, and $P(y_i - \hat{y}_i) = \mathbb{E}[|y_i - \hat{y}_i|^2]$ measures the noise power corresponding to the prediction error.

Finally, the overall loss of DAPKD-CFSL is expressed as the sum of the three parts

$$\mathcal{L}_{\mathrm{all}} = \mathcal{L}_{\mathrm{fsl}} + \mathcal{L}_{\mathrm{PKD}} + \mathcal{L}_{\mathrm{SNR}}. \tag{27}$$

In the meta-testing phase, the student model $\mathcal{N}_S$ optimized via the GL-PKD strategy will be employed to extract features from unlabeled samples in the target domain. Subsequently, these features will be classified using a K-NN classifier to generate predictions.

IV. Experiments

In this section, we provide a detailed introduction to the four HSI datasets used in the experiment and their various configurations. Additionally, we conducted ablation experiments and comparative experiments, and conducted an in-depth analysis of the results.

A. Dataset Description

1) Source Data: The Chikusei dataset used the Headwall Hyperspec-VNIR-CIRIS imaging spectrometer captures the spectral characteristics of mixed urban and rural landscapes in Ibaraki, Japan. This dataset comprises images of size 2517 × 2335 pixels, with 128 spectral bands ranging from 363 to 1018 nm and a spatial resolution of 2.5 m/pixel. It includes 19 feature classes representing both urban and rural areas. Fig. 4 shows pseudo-color composite images along with corresponding ground-truth labeled maps. Additionally, Table I provides a detailed description of the feature classes and the number of samples for each class.

2) Target Data: The Pavia University (PaviaU) dataset serves as a benchmark for HSI analysis and land cover classification research. Collected during a NASA/JPL airborne campaign using the AVIRIS sensor, this dataset covers an urban-academic landscape with spatial dimensions of 610 × 340 pixels. Each pixel contains calibrated reflectance values across 103 spectral bands spanning the visible-to-shortwave infrared spectrum, with water absorption bands systematically removed during preprocessing. The dataset categorizes nine distinct land cover types, such as oilseed rape fields, corn fields, and grasslands. A true-color composite ground-truth map with class-specific annotations is presented in Fig. 5, while Table II provides detailed statistics on pixel counts per land cover category.

The Salinas dataset, a benchmark resource for HSI analysis, was captured by the AVIRIS over agricultural regions in California's Salinas Valley. This dataset comprises a HSI with spatial dimensions of 512 × 217 pixels and 224 spectral bands spanning the visible to shortwave infrared spectrum. Featuring 16 distinct land cover categories, the SA dataset includes detailed classifications such as romaine lettuce fields, fallow soils, vineyard plots, and natural vegetation. A georeferenced ground-truth map with annotated class labels is provided in Fig. 6, while Table III quantitatively summarizes the pixel distribution across all agricultural and natural land cover categories.

The Houston dataset is a standard dataset used in HSI processing and urban land cover classification. Acquired by the CASI-1500 sensor during the 2013 IEEE GRSS Data Fusion Contest, it includes a 349 × 1905 pixel HSI with 144 spectral bands per pixel, covering wavelengths from 380 to 1050 nm. The dataset comprises 15 urban land cover classes, including healthy grass, asphalt roofs, roads, water bodies, and residential areas. The ground-truth map and color labels are shown in Fig. 7, and the pixel counts for each class are presented in Table IV.

B. Experimental Configuration

All relevant experiments were conducted on a uniform hardware platform consisting of an Intel Core i7-13400KF processor (3.80 GHz), 64 GB of RAM, and an NVIDIA GeForce RTX 4070 Ti GPU. The experimental environment was configured using Python 3.11 and PyTorch 2.3.0.

All samples were involved in training in the form of HSI blocks, with the image block size uniformly set to 9 × 9 pixels and the number of spectral channels fixed at $B = 100$.

In the pretraining stage, the teacher model with the same structure as the student model was used for parameter initialization. The training number of episodes is set to 200, 300, and 300 on the PaviaU, Salinas, and Houston datasets, respectively. The initial learning rate was uniformly set to $1e^{-3}$, the masking ratio was fixed at 0.5, and the noise levels were 150 (for the PaviaU and Salinas datasets) and 50 (for the Houston dataset). The above parameter configuration was consistent with the same settings of the DEMAE method [62], while the sensitivity analysis of the masking ratio and noise level would be further explored in subsequent ablation experiments.

During the meta-training stage, the teacher and student model architectures remain consistent. The meta-training tasks follow the C-way K-shot tasks requirements, with one sample per class in the support set and 19 samples per class in the query set to evaluate cross-domain generalization capability. The total number of training episodes is set to 5000, the

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TABLE V
ABLATION EXPERIMENTS THROUGH COMBINATIONS OF MODULES ON PAVIAU

Dataset	DB-SSA	GL-PKD	DA-PT		OA(%)	AA(%)	Kappa
			Diff	DA			
PaviaU	×	×	×	×	82.05±1.85	83.60±1.39	77.50±2.51
	✓	×	×	×	83.21±1.78	83.97±1.55	79.30±2.17
	×	✓	×	×	85.93±2.32	86.25±1.95	82.70±1.73
	✓	✓	×	×	87.10±2.10	87.77±2.11	84.98±2.46
	✓	✓	✓	×	88.02±1.92	88.64±1.89	86.21±2.14
	✓	✓	×	✓	87.74±2.43	88.01±2.29	86.53±1.96
	✓	✓	✓	✓	**88.64±2.02**	**89.31±2.34**	**88.29±1.37**

TABLE VI
ABLATION EXPERIMENTS THROUGH COMBINATIONS OF MODULES ON THE SALINAS

Dataset	DB-SSA	GL-PKD	DA-PT		OA(%)	AA(%)	Kappa
			Diff	DA			
Salinas	×	×	×	×	87.73±1.05	90.78±0.88	88.02±1.05
	✓	×	×	×	88.32±0.68	91.53±0.85	91.30±1.17
	×	✓	×	×	91.06±0.82	92.85±0.74	91.96±1.73
	✓	✓	×	×	91.88±0.70	94.48±0.81	93.10±0.96
	✓	✓	✓	×	92.52±1.01	94.95±0.80	93.45±1.23
	✓	✓	×	✓	92.39±0.82	95.12±0.69	93.74±1.06
	✓	✓	✓	✓	**93.48±0.62**	**96.62±0.67**	**94.43±1.18**

TABLE VII
ABLATION EXPERIMENTS THROUGH COMBINATIONS OF MODULES ON THE HOUSTON

Dataset	DB-SSA	GL-PKD	DA-PT		OA(%)	AA(%)	Kappa
			Diff	DA			
Houston	×	×	×	×	74.42±4.75	75.70±2.42	74.50±1.68
	✓	×	×	×	75.27±4.19	76.71±1.95	76.14±2.33
	×	✓	×	×	77.41±2.51	79.12±2.01	76.43±1.64
	✓	✓	×	×	78.95±1.85	80.53±1.70	79.39±2.52
	✓	✓	✓	×	80.69±2.60	82.13±1.64	81.02±1.83
	✓	✓	×	✓	80.05±3.26	81.45±2.27	80.62±2.49
	✓	✓	✓	✓	**81.81±1.63**	**83.72±3.80**	**81.74±3.46**

learning rate to 0.001, and the weight decay coefficient to 0.0001, which effectively suppresses overfitting and stabilizes network training.

To ensure the comparability of the evaluation results, all experiments adopted standard evaluation metrics: overall accuracy (OA), average accuracy (AA), and Kappa coefficient. The experimental results were calculated based on the arithmetic mean of ten independent experiments.

C. Ablation Experiment

To verify the effectiveness of the DAPKD-CFSL method and deeply analyze the role of the core modules, a sequential ablation experiment was designed. The experiment took the standard FSL model as the baseline. The DB-SSA module, GL-PKD module, and DA-PT module were incrementally incorporated into the framework. Specifically, during the evaluation of the DA-PT module, the complementary nature of the diffusion model (Diff) and DA training in achieving domain alignment was assessed by analyzing their individual and combined impacts. All experiments were trained based on five labeled samples. The results of the ablation experiments are presented in Tables V–VII.

The ablation experimental results from Tables V–VII indicate that, compared with the standard FSL framework, the addition of the DB-SSA module improves the classification performance by 1.16%, 0.59%, and 0.85%, respectively, effectively enhancing the expression ability of key features. Additionally, the GL-PKD module, through the collaborative effect of global prototype alignment and local contrastive learning, significantly improves the interclass separability and intraclass compactness, achieving OA improvements of 3.88%, 3.33%, and 2.99%. Notably, when the DB-SSA module is used in conjunction with the GL-PKD module, it further filters key features and strengthens the prototype alignment effect in the GL-PKD, resulting in additional OA improvements of 1.5%, 1.6%, and 2.0% on the three datasets. Finally, in the DA-PT module, the integration of diffusion enhancement and DA learning demonstrates superior performance compared to their individual applications, yielding OA increases of 0.6%, 0.96%, and 1.12% across the three datasets and mitigating domain-shift issues. Overall, the three modules proposed in this article are highly coordinated with the FSL framework, complementing each other, and can significantly improve the

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TABLE VIII

CLASSIFICATION ACCURACIES FOR DIFFERENT METHODS ON TARGET SCENES AND OVERALL CLASSIFICATION ACCURACY (%)—PAVIAU DATASET (FIVE LABELED SAMPLES FROM TD)

Class	Classification algorithms								
	SVM	SSRN	SPFormer	DEMAE	DCFSL	Gia-CFSL	SCFormer	GLGAT-CFSL	DAPKD-CFSL
1	77.82±1.23	54.12±1.89	68.58±2.12	70.15±1.56	86.68±1.45	74.72±2.34	85.73±1.67	**96.76±0.95**	95.78±1.78
2	73.10±1.45	69.42±2.01	81.58±1.78	86.94±1.34	89.10±1.89	92.03±1.23	85.29±1.56	85.98±2.12	**96.87±1.45**
3	46.81±2.34	66.09±1.67	52.94±1.89	74.15±2.01	74.70±1.23	79.50±1.56	58.42±1.78	79.46±1.45	**85.01±1.89**
4	62.25±1.56	78.05±1.23	87.30±1.45	92.50±1.89	90.67±2.12	86.50±1.78	85.16±1.67	**92.88±1.23**	91.84±1.56
5	93.33±1.89	92.63±1.45	96.41±1.23	98.85±0.14	97.03±1.78	99.60±0.39	99.58±0.32	97.74±1.89	**98.72±0.27**
6	35.20±2.12	63.16±1.56	72.17±1.89	87.84±1.23	74.64±1.45	75.75±1.78	75.23±1.67	**94.73±1.56**	92.49±1.89
7	45.13±1.78	79.36±1.23	73.41±1.45	79.54±1.89	83.14±1.56	91.55±1.67	85.07±1.23	90.08±1.45	**92.20±1.78**
8	70.63±1.45	57.64±1.89	58.64±1.23	80.99±1.56	52.61±1.78	73.40±1.67	80.35±1.89	72.64±1.23	**90.87±1.45**
9	98.27±0.72	97.39±1.56	97.39±1.89	98.10±0.89	98.82±0.17	98.17±0.82	98.91±0.15	93.54±1.89	**98.90±0.09**
OA	62.39±2.78	73.27±2.45	81.26±2.23	84.68±1.89	81.86±1.56	83.23±2.67	86.38±2.23	87.43±2.45	**88.64±2.02**
AA	66.47±1.89	71.78±2.23	84.25±1.56	85.84±1.78	82.18±2.45	82.98±1.67	84.52±2.23	88.60±2.89	**89.31±2.34**
Kappa	55.16±1.56	63.86±1.89	77.49±1.23	79.33±1.45	76.30±1.78	79.49±1.67	81.53±1.89	85.42±1.23	**88.29±1.37**

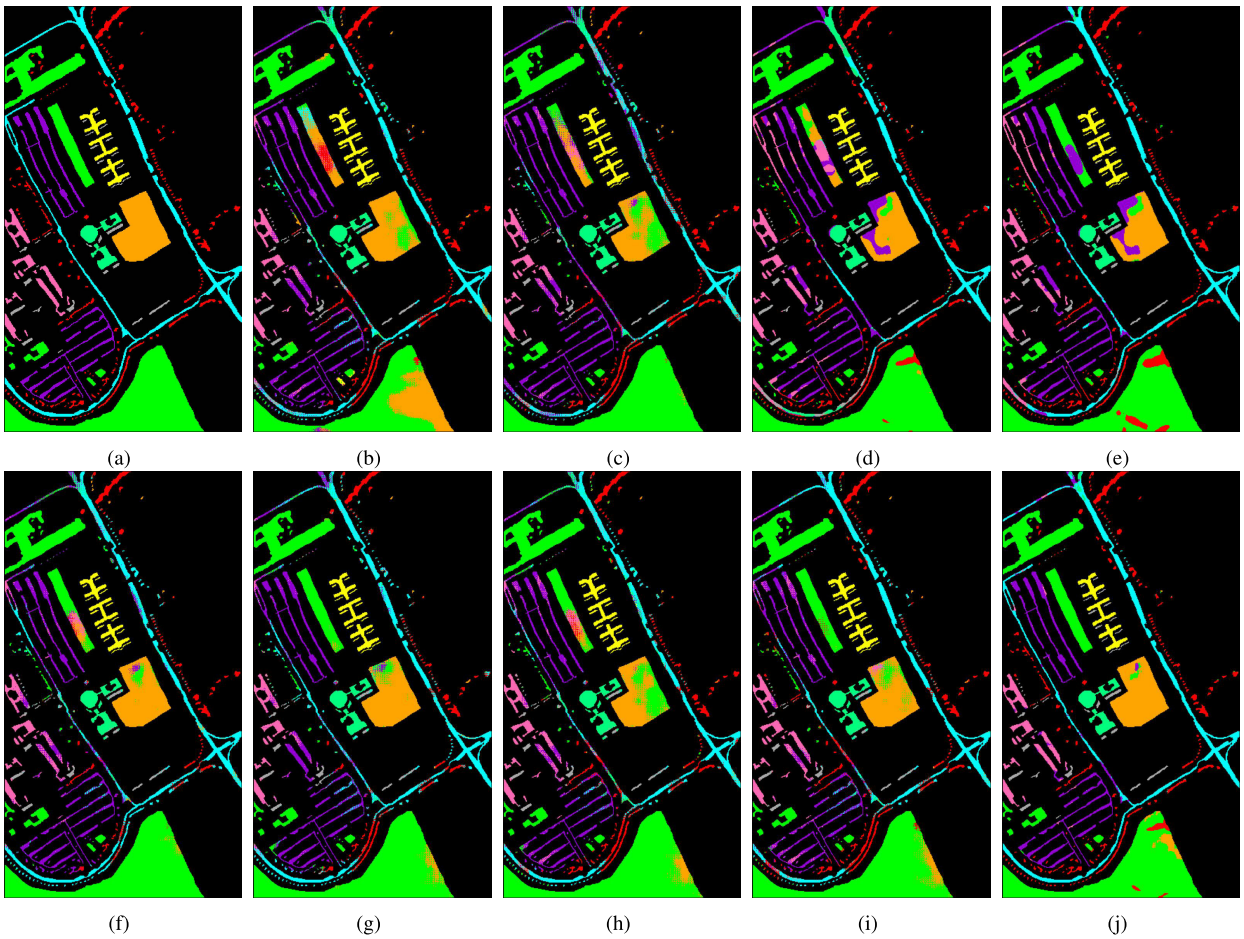


Fig. 8. PaviaU. (a) Ground-truth map. (b)–(j) Classification results for different methods. (b) SVM. (c) SSRN. (d) SPFormer. (e) DEMAE. (f) DCFSL. (g) Gia-CFSL. (h) SCFormer. (i) GLGAT-CFSL. (j) Our method.

efficiency and performance of FSL tasks with limited labeled samples.

D. Results of Comparative Experiments

To evaluate the effectiveness of the proposed DAPKD-CFSL method, a systematic comparison was performed against several state-of-the-art approaches, including SVM [7], SSRN [16], SPFormer [63], DAMAE [62], DCFSL [33], Gia-CFSL [34], SCFormer [39], and GLGAT-CFSL [36]. Among them, SVM serves as an ML-based approach, while SSRN and SPFormer represent DL-driven few-shot HSIC methods. Notably, DAMAE employs a novel self-supervised pretraining model. The latest approaches designed for CDFSL tasks include four advanced methods: DCFSL, Gia-CFSL, the Transformer-based SCFormer, and GLGAT-CFSL.

All experiments were conducted under identical conditions, with five labeled samples per class for training.

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TABLE IX
CLASSIFICATION ACCURACIES FOR DIFFERENT METHODS ON TARGET SCENES AND OVERALL CLASSIFICATION ACCURACY (%)—SALINAS DATASET (FIVE LABELED SAMPLES FROM TD)

Class	Classification algorithms								
	SVM	SSRN	SPFormer	DEMAE	DCFSL	Gia-CFSL	SCFormer	GLGAT-CFSL	DAPKD-CFSL
1	97.55±1.23	95.09±1.89	96.92±2.12	94.05±1.56	95.20±1.45	98.07±1.34	99.38±0.41	**99.74±0.15**	98.94±0.95
2	87.12±2.34	96.09±1.67	96.01±1.89	98.97±0.82	98.32±1.23	99.18±0.61	99.29±0.50	98.63±1.16	**99.41±0.48**
3	83.25±1.56	91.34±1.23	51.19±4.50	62.07±3.45	94.38±1.78	97.52±1.45	94.54±1.89	97.69±1.23	**99.51±0.38**
4	94.91±1.78	96.19±1.45	98.64±1.15	98.07±1.23	99.05±0.74	98.83±0.96	98.86±0.93	98.76±1.03	**99.30±0.59**
5	91.78±1.89	88.62±2.12	88.72±2.34	97.68±1.23	91.05±1.56	94.49±1.78	87.27±2.45	**98.08±1.71**	96.10±1.89
6	97.43±1.23	97.94±1.45	98.75±0.95	98.58±1.21	98.72±1.07	98.72±1.07	96.65±1.89	98.92±0.87	**99.34±0.55**
7	93.76±1.56	97.74±1.23	97.09±1.45	98.09±1.23	99.18±0.61	98.63±1.16	99.38±0.41	99.39±0.40	**99.56±0.33**
8	60.75±3.45	65.84±2.89	88.92±1.56	**90.76±1.23**	71.56±2.34	68.46±2.45	78.52±1.89	64.41±3.45	88.44±1.35
9	95.83±1.23	95.31±1.56	84.24±2.34	86.43±2.12	97.51±1.23	98.48±1.31	98.51±1.28	99.19±0.60	**99.74±0.15**
10	70.14±2.34	68.09±2.45	69.85±2.12	76.97±1.89	68.28±2.34	88.01±1.78	79.84±1.95	**92.62±1.17**	90.73±1.56
11	88.52±1.56	91.39±1.23	49.31±4.50	**99.34±0.55**	98.78±0.01	97.78±1.01	98.67±1.12	97.74±1.05	96.97±1.82
12	96.72±1.23	98.68±0.11	93.03±1.56	71.19±2.34	97.82±1.97	99.34±0.55	99.40±0.49	94.47±1.32	**99.64±0.25**
13	97.18±1.23	93.16±1.56	98.45±1.34	98.69±1.10	98.30±1.49	98.15±1.64	99.04±0.75	99.26±0.53	**99.47±0.42**
14	91.29±1.56	94.42±1.23	98.66±1.13	72.83±2.34	98.85±0.94	98.89±0.90	99.02±0.77	99.56±0.23	**99.74±0.15**
15	61.75±3.45	69.03±2.34	56.74±3.45	63.79±2.34	85.83±1.56	79.89±1.89	77.33±2.45	85.22±1.56	**86.16±1.63**
16	85.22±1.56	80.91±2.34	83.66±1.56	73.28±2.34	88.41±1.38	**95.02±1.77**	85.09±1.70	94.65±1.14	90.20±1.59
OA	80.25±1.56	82.62±1.15	89.74±1.26	92.21±1.64	89.14±1.85	89.82±1.46	92.15±1.82	93.05±1.71	**93.48±0.62**
AA	88.06±1.12	88.72±1.63	90.60±1.04	94.57±1.83	92.26±1.39	93.18±1.61	93.96±1.48	95.19±1.72	**96.62±0.67**
Kappa	82.05±1.56	82.80±1.23	90.42±1.45	92.63±1.23	88.31±1.56	88.74±1.23	92.63±1.56	93.68±1.34	**94.43±1.18**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 9. Salinas. (a) Ground-truth map. (b)–(j) Classification results for different methods. (b) SVM. (c) SSRN. (d) SPFormer. (e) DEMAE. (f) DCFSL. (g) Gia-CFSL. (h) SCFormer. (i) GLGAT-CFSL. (j) Our method.

Noncross-domain few-shot methods, including SVM, SSRN, SPFormer, and DEMAE, were directly trained and tested within the domains of the PaviaU, Salinas, and Houston datasets. The pretraining and data augmentation settings in DEMAE were kept consistent with those in our method. Cross-domain methods, including DCFSL, Gia-CFSL, SCFormer, GLGAT-CFSL, and ours, utilized the Chikusei dataset as the source domain and one of the PaviaU, Salinas, or Houston datasets as the target domain. The settings for the meta-tasks were consistently maintained across all methods. All other hyperparameter configurations followed the original implementations described in their respective references. To minimize the influence of randomness on the experimental outcomes, all methods were independently repeated ten times using the same set of random samples. The classification results for each model are summarized in Tables VIII–X, with the best performance for each evaluation metric highlighted in bold.

1) Experimental Result of PaviaU Dataset: As shown in Table VIII, the experimental data on the classification accuracy of the PaviaU dataset indicates that our method demonstrates significant advantages over existing benchmark models, achieving performance improvements of 1.21%, 1.58%, and 1.58% better than the second-best method in OA, AA, and Kappa, respectively. The visualization results, which is shown in Fig. 8, further verify that our method maintains a leading edge in both fine-grained and macroscopic classification tasks, particularly showing outstanding classification consistency in the fine-grained feature areas marked in purple and red and the macroscopic structure areas marked in brown.

2) Experimental Result of Salinas Dataset: Table IX shows the details of the classification accuracy achieved by the comparison method on the Salinas dataset. The experimental results show that our method achieves the best classification accuracy in terms of OA, AA, and Kappa, reaching 93.48%, 96.62%, and 94.43%, respectively. Our method maintains a leading advantage in 10 out of 16 land cover classes and is significantly superior to other comparison algorithms. Fig. 9 further verifies that DAPKD-CFSL not only accurately distinguishes the boundaries of adjacent land covers, but also demonstrates better classification effects on large-scale land covers.

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TABLE X
CLASSIFICATION ACCURACIES FOR DIFFERENT METHODS ON TARGET SCENES AND OVERALL CLASSIFICATION ACCURACY (%)—HOUSTON DATASET (FIVE LABELED SAMPLES FROM TD)

Class	Classification algorithms								
	SVM	SSRN	SPFormer	DEMAE	DCFSL	Gia-CFSL	SCFormer	GLGAT-CFSL	DAPKD-CFSL
1	68.57±9.17	86.52±7.70	85.69±7.35	88.92±6.40	73.73±6.00	90.26±6.81	91.42±6.14	**90.88±8.16**	81.56±9.07
2	77.25±5.13	84.88±6.66	86.58±6.16	79.03±4.23	85.36±5.90	83.51±9.90	74.23±4.96	72.77±9.91	**87.85±6.30**
3	90.67±2.24	86.01±7.12	97.35±0.98	96.53±3.19	96.49±2.45	94.61±8.24	87.31±8.42	97.77±7.36	**98.53±1.66**
4	81.81±7.16	90.96±2.28	91.34±5.78	**91.79±3.21**	82.21±5.02	93.52±2.22	84.89±2.96	87.02±6.36	91.24±4.13
5	89.28±8.74	90.49±7.72	88.24±7.92	95.24±3.80	94.02±7.78	98.56±1.65	91.68±5.91	95.97±5.72	**98.58±3.66**
6	77.21±7.00	77.94±5.67	90.00±5.29	80.97±6.58	85.97±7.41	87.62±5.63	70.27±6.22	81.53±4.46	**91.19±5.97**
7	65.35±9.10	63.49±7.86	72.38±9.24	73.61±8.70	63.06±3.90	64.37±4.48	61.40±12.61	70.80±7.75	**74.70±5.04**
8	34.88±9.66	39.59±8.77	47.98±7.86	48.67±5.01	42.82±4.57	47.94±9.50	43.13±9.51	50.44±7.57	**51.09±6.44**
9	44.30±7.42	59.05±9.02	51.11±4.47	63.18±3.96	55.17±7.35	58.27±5.48	**79.00±3.36**	67.46±7.85	73.20±6.49
10	47.61±9.97	63.86±7.38	64.79±18.13	**75.14±4.98**	66.04±5.39	65.87±8.96	65.25±7.99	71.32±4.21	75.09±6.17
11	54.02±6.41	56.19±9.69	64.18±4.03	79.93±8.75	65.86±5.44	74.20±5.35	79.42±8.48	79.18±9.61	**82.93±7.43**
12	52.93±5.99	55.81±6.07	60.78±9.47	**73.66±7.54**	65.22±7.01	55.12±5.25	47.17±7.92	54.86±8.81	57.60±7.32
13	54.04±4.29	54.87±6.20	85.60±1.47	90.91±3.36	86.66±7.66	90.88±5.72	29.59±3.65	**95.73±0.94**	87.00±8.83
14	79.79±0.56	80.29±5.21	96.45±3.91	**100.00±0.00**	88.30±8.13	99.49±0.54	87.90±4.50	99.98±0.07	**100.00±0.00**
15	85.26±5.71	97.15±2.50	99.03±0.55	95.73±2.57	97.85±5.01	99.90±0.02	92.89±2.01	**100.00±0.00**	97.91±3.54
OA	67.09±3.58	74.33±2.44	76.93±2.72	80.38±2.67	75.54±2.14	78.43±1.45	80.80±1.94	80.04±1.70	**81.81±1.63**
AA	73.55±3.53	77.40±1.81	78.14±2.22	81.38±2.41	79.08±1.68	80.64±1.39	80.92±1.46	81.29±1.47	**83.72±3.80**
Kappa	64.49±3.85	72.28±2.63	77.87±2.93	79.97±2.88	75.58±2.31	76.70±1.57	79.37±2.08	80.07±1.85	**81.74±3.46**

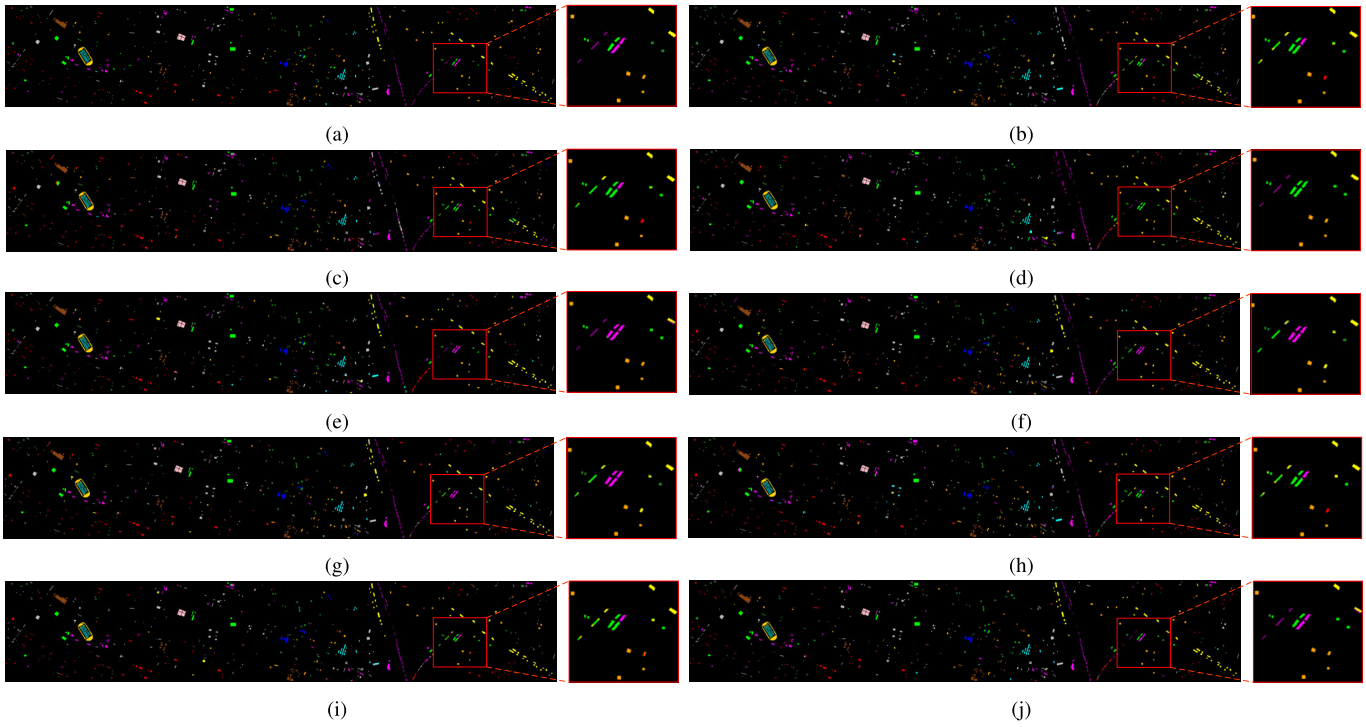


Fig. 10. Houston. (a) Ground-truth map. (b)–(j) Classification results for different methods. (b) SVM. (c) SSRN. (d) SPFormer. (e) DEMAE. (f) DCFSL. (g) Gia-CFSL. (h) SCFormer. (i) GLGAT-CFSL. (j) Our method.

3) Experimental Result of Houston Dataset: The classification accuracy comparison results of the Houston dataset presented in Table X indicate that our method achieves classification performance improvements of 1.77%, 2.43%, and 1.67% in terms of OA, AA, and Kappa, respectively, compared to the existing suboptimal methods. As illustrated in the visual comparison analysis in Fig. 10, our method generates more refined classification results for the categories in the right half of the shaded area compared to most of the comparison models. This is attributed to the fact that our method can extract key features from samples with different types of masks and noise during the training stage. Therefore, during the classification stage, our method does not encounter the problem of category misjudgment caused by shadow interference.

V. DISCUSSION

A. Different Numbers of Training Samples

To systematically explore the influence of the numbers of training samples on the performance of the model, Fig. 11

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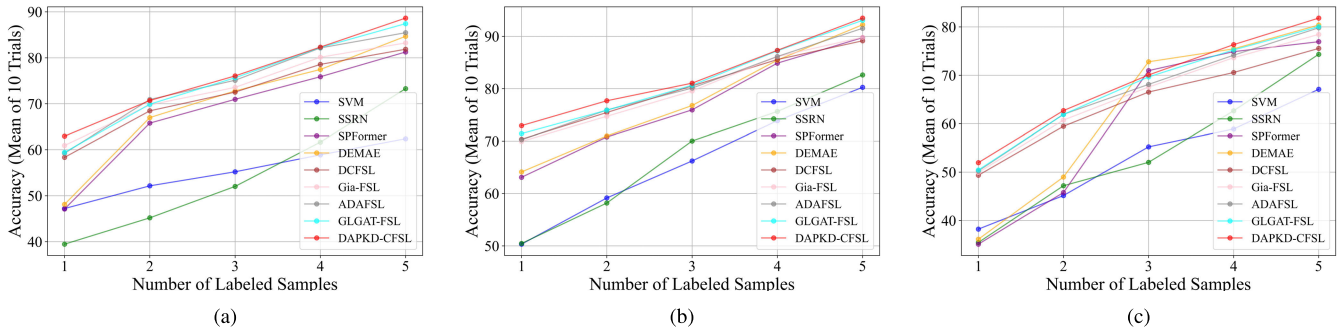


Fig. 11. Effect of different numbers of training samples on the performance of different methods on three target-domain datasets. (a) PaviaU. (b) Salinas. (c) Houston.

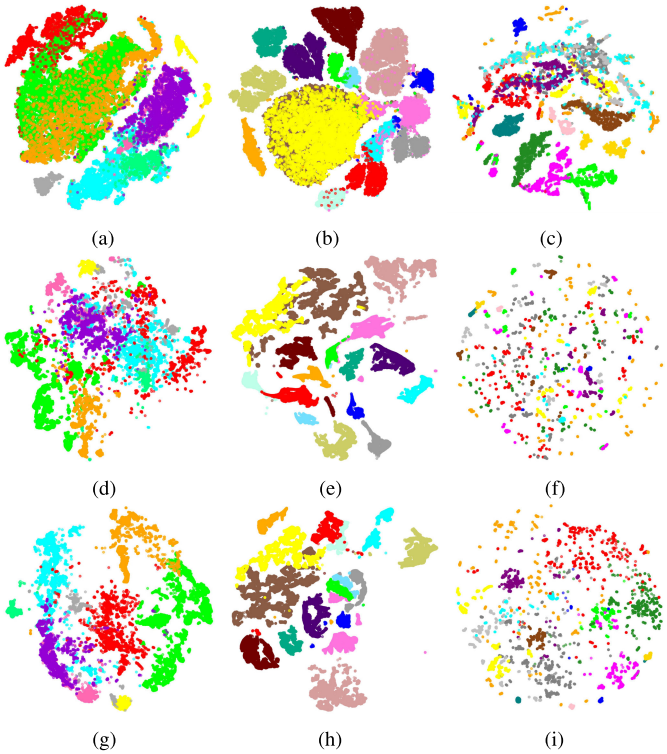


Fig. 12. t-SNE visualizations compare raw data and processed features from three datasets. (a) Original data of PaviaU. (b) Original data of Salinas. (c) Original data of Houston. (d) DB-SSA processed features of PaviaU. (e) DB-SSA processed features of Salinas. (f) DB-SSA processed features of Houston. (g) DAPKD-CFSL processed features of PaviaU. (h) DAPKD-CF SL processed features of Salinas. (i) DAPKD-CFSL processed features of Houston.

presents a comparison of the classification performance of different methods under different training labeled sample number ($L = \{1, 2, 3, 4, 5\}$) through a controlled variable experiment.

As shown in Fig. 11, moderately increasing the number of FSL training samples can significantly enhance the classification accuracy and generalization ability of the model. Specifically, when the sample size is small ($L < 3$), the model is prone to overfitting; however, when the sample size increases to $L = 5$, the TD classification accuracy reaches its highest level, and each category can achieve stable convergence with only limited supervised information. Through comprehensive performance evaluation, this study determines $L = 5$ as the optimal sample configuration scheme. This scheme not only fully captures the category feature representation, but also effectively suppresses the risk of overfitting. Notably, in all experimental groups of the PaviaU, Salinas, and Houston datasets, the DAPKD-CFSL method consistently demonstrates the highest classification accuracy and the best stability indicators.

B. Feature Separability

The t-distributed stochastic neighbor embedding (t-SNE) algorithm is utilized to conduct dimensionality reduction and visualization for both the original data and the high-dimensional features extracted by DAPKD-CFSL. The classification performance of the model is visually evaluated through 2-D feature distribution maps. Fig. 12(a)–(c) depicts the original data distributions of the PaviaU, Salinas, and Houston datasets, whereas Fig. 12(d)–(f) presents the feature space distributions after processing with DB-SSA module and Fig. 12(g)–(i) presents the feature space distributions after processing with DAPKD-CFSL.

In the PaviaU dataset, the original data exhibit significant dispersion and sample overlap in the feature space, leading to blurred category boundaries. The DB-SSA module focuses on category edges and regions of structural mutation, suppressing redundant noise interference. After further optimization using the DAPKD-CFSL framework, samples within the same category become more compact, while clear boundaries emerge between different categories. Notably, as shown in Fig. 12(d) and (g), after processing by the DB-SSA module, the categories represented by yellow, red, and green have initially shown a trend of separation. Subsequent DAPKD-CFSL processing further enhances the separability of these categories, forming highly distinct distribution areas.

For the Salinas dataset, while the original distribution demonstrates a multicenter aggregation pattern, it suffers from issues such as large intraclass feature dispersion and blurred interclass boundaries. The DB-SSA module achieves initial feature purification through spectral screening and spatial focusing, and DAPKD-CFSL further extracts features that significantly reduce intraclass distances while increasing interclass distances. Specifically, as shown in Fig. 12(e) and (h), the DB-SSA module initially separates the mixed area between the yellow and brown categories, while DAPKD-CFSL further

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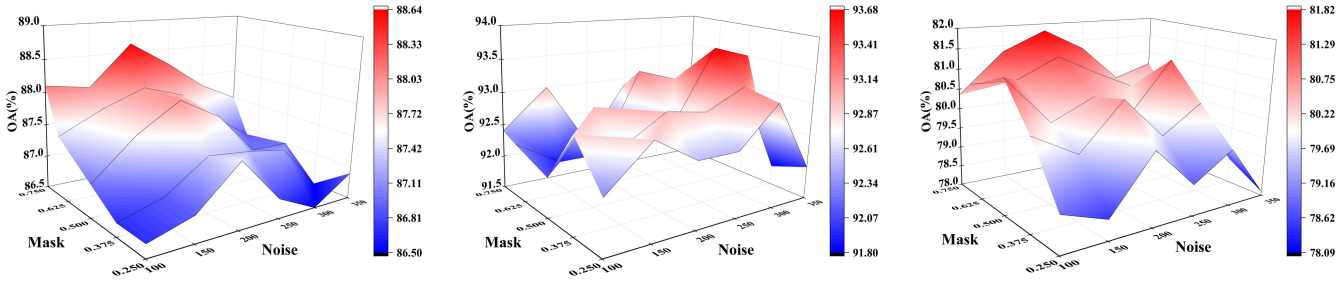


Fig. 13. Hyperparameter tuning on the performance of the proposed DAPKD-CFSL on three target-domain datasets. (a) PaviaU. (b) Salinas. (c) Houston.

TABLE XI
TRAINING TIME, TESTING TIME, FLOPS, AND PARAMETERS ON EACH DATASET WITH DIFFERENT METHODS

	Methods	DEMAE	DCFSL	Gia-CFSL	SCFormer	GLGAT-CFSL	DAPKD-CFSL
PaviaU	Training time	2282.50s	1575.24s	3034.27s	2034.27s	1942.04s	2178.50s
	Testing time	5.26s	9.01s	6.98s	4.08s	4.05s	3.85s
	FLOPs	1.8441G	1.5424G	2.5978G	1.9325G	2.1236G	2.3218G
	Parameters	4.039M	4.442M	4.882M	4.851M	5.924M	4.240M
Salina	Training time	3092.78s	2445.06s	3605.06s	2431.06s	2996.35s	3008.06s
	Testing time	7.24s	13.06s	9.87s	5.87s	5.46s	5.61s
	FLOPs	10.2441G	11.0424G	11.5978G	10.3325G	12.3236G	11.6315G
	Parameters	0.853M	0.524M	0.648M	0.685M	0.782M	0.645M
Houston	Training time	2410.78s	1986.78s	2605.06s	1959.06s	2196.35s	2360.10s
	Testing time	7.24s	13.06s	9.87s	5.87s	5.46s	5.16s
	FLOPs	2.2441G	2.0424G	2.5978G	1.9325G	2.3236G	2.1236G
	Parameters	4.619M	4.024M	4.848M	4.185M	4.184M	4.012M

realizes the spatial decoupling of the two types of samples, validating the model's powerful feature extraction capability.

Regarding the Houston dataset, characterized by extremely imbalanced sample sizes and shadow region overlaps between classes, the DB-SSA module provides noise suppression and feature enhancement for subsequent optimization through spectral–spatial filtering. The DAPKD-CFSL demonstrates a strong feature extraction capability. The feature visualization results in Fig. 12(f) and (i) indicate that the DB-SSA module initially alleviates the distribution skew caused by sample imbalance, and the DAPKD-CFSL not only enhances intra-class aggregation for minority classes, but also successfully alleviates interclass confusion caused by shadow interference, thereby significantly improving feature separability.

C. Hyperparameter Analysis

This section focuses on analyzing the two hyperparameters: noise and masking. Since our method is primarily designed for classification tasks rather than generative tasks, it is unnecessary to strictly enforce that high-noise-level data conform to a standard normal distribution. During the diffusion-augmented pretraining (DA-PT) stage, we set the noise level within a medium range [100, 150, 200, 250, 300, 350] and adopt HSI patch-masking ratios of [0.25, 0.375, 0.5, 0.675, 0.75], thereby introducing controllable noise interference while preserving the original spectral features. In the meta-training stage, both the noise intensity and the masking ratio are reduced to 50% of their respective values in the DA-PT stage.

The comparison of OA in Fig. 13 shows that when the noise level and masking ratio are set to 200 and 0.75, respectively, the PaviaU and Houston datasets achieve the best classification performance; whereas when set to 300 and 0.5, the model achieves optimal classification performance on the Salinas dataset. We attribute the variations in classification performance caused by these parameter differences primarily to the distinct scene characteristics: in urban scenes (e.g., PaviaU and Houston), where spectral differences among buildings, roads, and other features are pronounced, a moderate noise level suffices to guide the model to capture discriminative features. In contrast, in farmland scenes (e.g., Salinas), where spectral similarities among crops are high, enhanced noise interference is required to force the model to explore subtle feature differences, thereby improving class separability.

D. Computational Complexity

To assess the computational efficiency of FSL-based methods, we systematically evaluated the computational efficiency of each method across four dimensions: training time, testing time, parameter count, and floating-point operations (FLOPs) in Table XI. The experiments were conducted using a target-domain dataset with five labeled samples per category. In the training phase, the time cost primarily involved pretraining and FSL, while the testing phase encompassed inference computations for all HSI data.

Table XI shows that although DAPKD-CFSL exhibits slightly higher training time and FLOPs compared to other methods due to its diffusion model architecture, it achieves

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outstanding classification accuracy with fewer total parameters. Additionally, compared with the similar pretraining method DEMAE, DAPKD-CFSL demonstrates a faster training speed.

VI. Conclusion

This study proposes a cross-domain few-shot HSIC method based on diffusion-augmented prototype knowledge distillation. It achieves coordinated and efficient knowledge transfer through the GL-PKD strategy. By integrating prototype learning and contrastive learning mechanisms, it adjusts the intradomain class distribution differences to guide the student model to capture more representative knowledge representations, thereby achieving global feature matching. Additionally, to overcome the impact of domain shift and noise interference on domain alignment, a DA-PT framework is constructed. Through the joint denoising reconstruction task and DA constraints, it uses source-domain samples and unlabeled target-domain samples to co-train the teacher network, effectively extracting robust cross-domain features and focusing on mining domain-invariant features to suppress domain shift. Moreover, the proposed DB-SSA module significantly enhances the feature expression ability of discriminative spectral bands and key spatial regions by capturing global dependencies in the spectral dimension and local correlations in the spatial dimension. Comprehensive and extensive experiments and comparisons on three datasets show that the DAPKD-CFSL method has outstanding classification performance under limited sample conditions.

To the best of authors' knowledge, this study is an early attempt to introduce the knowledge distillation strategy in the CDFSL classification task. Future work will focus on exploring more efficient teacher model pretraining methods and nonoverlapping distillation frameworks in open-set CDFSL scenarios. In addition, a comprehensive investigation of the impact of architectural differences between teacher and student models on distillation effects will be conducted to further enhance the generalization ability and applicability of the model.

References

[1] D. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.

[2] P. Ghamisi et al., "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[3] I. Makki, R. Younes, C. Francis, T. Bianchi, and M. Zucchetti, "A survey of landmine detection using hyperspectral imaging," *ISPRS J. Photogramm. Remote Sens.*, vol. 124, pp. 40–53, Feb. 2017.

[4] S. Peyghambari and Y. Zhang, "Hyperspectral remote sensing in lithological mapping, mineral exploration, and environmental geology: An updated review," *J. Appl. Remote Sens.*, vol. 15, no. 3, Jul. 2021, Art. no. 031501.

[5] L. Liang et al., "Estimation of crop LAI using hyperspectral vegetation indices and a hybrid inversion method," *Remote Sens. Environ.*, vol. 165, pp. 123–134, Aug. 2015.

[6] M. Shimoni, R. Haelterman, and C. Perneel, "Hyperspectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag. (Replaces Newslett.)*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

[7] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[8] L. Samaniego, A. Bardossy, and K. Schulz, "Supervised classification of remotely sensed imagery using a modified k-NN technique," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2112–2125, Jul. 2008.

[9] Y. Ren, Y. Zhang, W. Wei, and L. Li, "A spectral–spatial hyperspectral data classification approach using random forest with label constraints," in *Proc. IEEE Workshop Electron., Comput. Appl.*, May 2014, pp. 344–347.

[10] J. A. Benediktsson, J. H. Palmason, and J. S. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.

[11] T. V. Bandos, L. Bruzzone, and G. Camps-Valls, "Classification of hyperspectral images with regularized linear discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 3, pp. 862–873, Mar. 2009.

[12] P. Wang, H. Yao, C. Li, G. Zhang, and H. Leung, "Multiresolution analysis based on dual-scale regression for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5406319.

[13] Y. Zhang, C. Wu, T. Zhang, Y. Liu, and Y. Zheng, "Self-attention guidance and multiscale feature fusion-based UAV image object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[14] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[15] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[16] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[17] Y. Xu, L. Zhang, B. Du, and F. Zhang, "Spectral–spatial unified networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5893–5909, Oct. 2018.

[18] S. Mei, J. Ji, Y. Geng, Z. Zhang, X. Li, and Q. Du, "Unsupervised spatial–spectral feature learning by 3D convolutional autoencoder for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6808–6820, Sep. 2019.

[19] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[20] B. Xi et al., "Semisupervised cross-scale graph prototypical network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 9337–9351, Nov. 2023.

[21] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[22] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[23] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014.

[24] W. Li, C. Chen, M. Zhang, H. Li, and Q. Du, "Data augmentation for hyperspectral image classification with deep CNN," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 4, pp. 593–597, Apr. 2019.

[25] J. M. Haut, M. E. Paoletti, J. Plaza, A. Plaza, and J. Li, "Hyperspectral image classification using random occlusion data augmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 11, pp. 1751–1755, Nov. 2019.

[26] C. Ding et al., "Integrating prototype learning with graph convolution network for effective active hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504816.

[27] H. Wu and S. Prasad, "Semi-supervised deep learning using pseudo labels for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1259–1270, Mar. 2018.

[28] N. Wambugu et al., "Hyperspectral image classification on insufficient-sample and feature learning using deep neural networks: A review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 105, Dec. 2021, Art. no. 102603.

[29] C. Zhang, J. Yue, and Q. Qin, "Global prototypical network for few-shot hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4748–4759, 2020.

[30] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

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[31] W. Wang, L. Ma, M. Chen, and Q. Du, "Joint correlation alignment-based graph neural network for domain adaptation of multitemporal hyperspectral remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3170–3184, 2021.
[32] C. Zhao, B. Qin, S. Feng, W. Zhu, L. Zhang, and J. Ren, "An unsupervised domain adaptation method towards multi-level features and decision boundaries for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5546216.
[33] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.
[34] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 1–14, Feb. 2024.
[35] F. Sun, H. Wu, Z. Luo, W. Gu, Y. Yan, and Q. Du, "Informative feature selection for domain adaptation," *IEEE Access*, vol. 7, pp. 142551–142563, 2019.
[36] C. Ding et al., "GLGAT-CFSL: Global-local graph attention network-based cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522519.
[37] B. Qin, S. Feng, C. Zhao, W. Li, R. Tao, and W. Xiang, "Cross-domain few-shot learning based on feature disentanglement for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5514215.
[38] Q. Liu et al., "Refined prototypical contrastive learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506214.
[39] J. Li, Z. Zhang, R. Song, Y. Li, and Q. Du, "SCFormer: Spectral coordinate transformer for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 33, pp. 840–855, 2024.
[40] Z. Ye, J. Wang, H. Liu, Y. Zhang, and W. Li, "Adaptive domain-adversarial few-shot learning for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5532017.
[41] C. Fu, B. Du, and L. Zhang, "ReSC-net: Hyperspectral image classification based on attention-enhanced residual module and spatial-channel attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5518615.
[42] Y. Dang, H. Li, B. Liu, and X. Zhang, "Cross-domain few-shot learning for hyperspectral image classification based on global-to-local enhanced channel attention," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.
[43] H. Wu, Z. Xue, S. Zhou, and H. Su, "Beyond spectral shift mitigation: Knowledge swap net for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5528218.
[44] C. Yu et al., "Distillation-constrained prototype representation network for hyperspectral image incremental classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5507414.
[45] S. Feng, H. Zhang, B. Xi, C. Zhao, Y. Li, and J. Chanussot, "Cross-domain few-shot learning based on decoupled knowledge distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534414.
[46] Z. Fang et al., "Confident learning-based domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527116.
[47] Y. Huang et al., "Two-branch attention adversarial domain adaptation network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540813.
[48] Z. Fang, W. He, Z. Li, Q. Du, and Q. Chen, "Masked self-distillation domain adaptation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5525720.
[49] Y. Xia, L.-J. Yun, and C. Yang, "Transferable adversarial masked self-distillation for unsupervised domain adaptation," *Complex Intell. Syst.*, vol. 9, no. 6, pp. 6567–6580, Dec. 2023.
[50] J. Peng, Y. Huang, W. Sun, N. Chen, Y. Ning, and Q. Du, "Domain adaptation in remote sensing image classification: A survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9842–9859, 2022.
[51] S. Deng, D. Liao, X. Gao, J. Zhao, and K. Ye, "A survey on cross-domain few-shot image classification," in *Proc. Int. Conf. Big Data*, Honolulu, HI, USA. Berlin, Germany: Springer, 2023, pp. 3–17.
[52] Y. Zhang, W. Li, R. Tao, J. Peng, Q. Du, and Z. Cai, "Cross-scene hyperspectral image classification with discriminative cooperative alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9646–9660, Nov. 2021.
[53] S. K. Meher and N. S. Kothari, "Interpretable rule-based fuzzy ELM and domain adaptation for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5907–5919, Jul. 2021.
[54] W. Wang, F. Liu, J. Liu, and L. Xiao, "Cross-domain few-shot hyperspectral image classification with class-wise attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502418.
[55] J. Gou, B. Yu, S. J. Maybank, and D. Tao, "Knowledge distillation: A survey," *Int. J. Comput. Vis.*, vol. 129, no. 6, pp. 1789–1819, Jun. 2021.
[56] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.
[57] J. Yue, L. Fang, H. Rahmani, and P. Ghamisi, "Self-supervised learning with adaptive distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501813.
[58] W. Zhao, R. Peng, Q. Wang, C. Cheng, W. J. Emery, and L. Zhang, "Life-long learning with continual spectral–spatial feature distillation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5544214.
[59] Q. Chi, G. Lv, G. Zhao, and X. Dong, "A novel knowledge distillation method for self-supervised hyperspectral image classification," *Remote Sens.*, vol. 14, no. 18, p. 4523, 2022.
[60] Y. Zhang et al., "Ensemble network-based distillation for hyperspectral image classification in the presence of label noise," *Remote Sens.*, vol. 16, no. 22, p. 4247, Nov. 2024.
[61] K. Preechakul, N. Chatthee, S. Wizadwongsa, and S. Suwajanakorn, "Diffusion autoencoders: Toward a meaningful and decodable representation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10619–10629.
[62] Z. Li et al., "DEMAE: Diffusion-enhanced masked autoencoder for hyperspectral image classification with few labeled samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5527616.
[63] Z. Li, Z. Xue, Q. Xu, L. Zhang, T. Zhu, and M. Zhang, "SPFormer: Self-pooling transformer for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502019.



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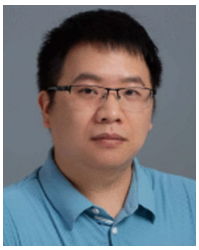
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